WORKED SOLUTION

STEP III 2011, Question 3

Pure • Factorisation

STEP III 2011, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q3

The key idea

Two cubes $(x-\alpha)^3$ and $(x-\beta)^3$ give us two free constants a and b, and a monic cubic with no x^2 term imposes four conditions. So we use two of the conditions to find a and b, and the other two must then be exactly the facts about $\alpha+\beta$ and $\alpha\beta$ that the quadratic supplies. Once the cubic is a sum of two cubes, setting it to zero becomes "a ratio cubed equals a constant", which we solve by taking cube roots, all three of them.

Solution

Writing the cubic as a sum of two cubes

Since $p \neq 0$, the quadratic $pt^2 - qt + p^2 = 0$ has

$$\alpha+\beta = \frac{q}{p}, \qquad \alpha\beta = \frac{p^2}{p} = p,$$

and its discriminant is $q^2 - 4p^3$. The proviso $q^2 \neq 4p^3$ therefore says precisely that $\alpha \neq \beta$. Also $\alpha\beta = p \neq 0$, so neither root is zero.

Expanding,

$$a(x-\alpha)^3 + b(x-\beta)^3 = (a+b)x^3 - 3(a\alpha+b\beta)x^2 + 3(a\alpha^2+b\beta^2)x - (a\alpha^3+b\beta^3).$$

To match $x^3 - 3px + q$ we need

$$a+b=1, \qquad a\alpha+b\beta=0, \qquad a\alpha^2+b\beta^2=-p, \qquad a\alpha^3+b\beta^3=-q.$$

The first two are linear in a and b. Substituting $b = 1-a$ into the second gives $a(\alpha-\beta) = -\beta$, and since $\alpha \neq \beta$ we may divide:

$$a = \frac{\beta}{\beta-\alpha}, \qquad b = \frac{\alpha}{\alpha-\beta}.$$

Now we check that the remaining two conditions hold automatically:

$$a\alpha^2 + b\beta^2 = \frac{\beta\alpha^2 - \alpha\beta^2}{\beta - \alpha} = \frac{\alpha\beta(\alpha - \beta)}{\beta - \alpha} = -\alpha\beta = -p,$$
$$a\alpha^3 + b\beta^3 = \frac{\beta\alpha^3 - \alpha\beta^3}{\beta - \alpha} = \frac{\alpha\beta(\alpha - \beta)(\alpha + \beta)}{\beta - \alpha} = -\alpha\beta(\alpha + \beta) = -p \cdot \frac{q}{p} = -q.$$

So, with these values of a and b,

$$x^3 - 3px + q = \frac{\beta}{\beta - \alpha}(x - \alpha)^3 + \frac{\alpha}{\alpha - \beta}(x - \beta)^3,$$

as required.

Solving $x^3 - 24x + 48 = 0$

Here $p = 8$ and $q = 48$, and $q^2 = 2304 \neq 2048 = 4p^3$, so the result applies. The quadratic is $8t^2 - 48t + 64 = 0$, that is $t^2 - 6t + 8 = 0$, with roots 2 and 4. Take $\alpha = 2$, $\beta = 4$; then $a = 2$ and $b = -1$, so

$$x^3 - 24x + 48 = 2(x - 2)^3 - (x - 4)^3.$$

(A quick expansion confirms this: $2x^3 - 12x^2 + 24x - 16 - x^3 + 12x^2 - 48x + 64$.)

Ask yourself How do we turn $2(x - 2)^3 = (x - 4)^3$ into something we can solve?

Divide by $(x - 4)^3$. This is allowed, because $x = 4$ is not a root ($64 - 96 + 48 = 16$). The equation becomes

$$\left(\frac{x - 2}{x - 4}\right)^3 = \frac{1}{2}.$$

Write $c = 2^{1/3}$. The complex numbers whose cube is $\frac{1}{2}$ are $k = c^{-1}$, $c^{-1}\omega$ and $c^{-1}\omega^2$. For each, solve $\dfrac{x - 2}{x - 4} = k$:

$$x - 2 = kx - 4k \quad \Longrightarrow \quad x = \frac{2 - 4k}{1 - k},$$

which is valid since $k \neq 1$. The map $k \mapsto x$ is one-to-one, so the three values of k give three distinct roots, which must be all the roots of the cubic.

For $k = c^{-1}$, multiply top and bottom by c:

$$x = \frac{2c - 4}{c - 1} = \frac{2(2 - 2^{1/3})}{1 - 2^{1/3}},$$

which is the stated solution. For $k = c^{-1}\omega$, multiply top and bottom by $c\,\omega^{-1} = c\,\omega^2$ (using $\omega^3 = 1$):

$$x = \frac{2c\omega^2 - 4}{c\omega^2 - 1} = \frac{2(2 - 2^{1/3}\omega^2)}{1 - 2^{1/3}\omega^2},$$

and in the same way $k = c^{-1}\omega^2$ gives

$$x = \frac{2(2 - 2^{1/3}\omega)}{1 - 2^{1/3}\omega}.$$

So the three solutions are $x = \dfrac{2(2 - 2^{1/3}\omega^j)}{1 - 2^{1/3}\omega^j}$ for $j = 0, 1, 2$.

The case $p = r^2$, $q = 2r^3$

Now $q^2 = 4r^6 = 4p^3$, so this is exactly the case the method excludes: the quadratic $r^2t^2 - 2r^3t + r^4 = r^2(t - r)^2$ has the repeated root $t = r$, and we cannot find a and b. But a repeated root of the quadratic suggests a repeated root of the cubic too. Indeed $x = r$ gives $r^3 - 3r^3 + 2r^3 = 0$, and the derivative $3x^2 - 3r^2$ also vanishes at $x = r$, so $(x - r)^2$ is a factor. Comparing the x^3 and constant terms,

$$x^3 - 3r^2x + 2r^3 = (x - r)^2(x + 2r).$$

The roots are $x = r$ (twice) and $x = -2r$.

? **Ask yourself** Could we have found these from the earlier formula?

No. With $\alpha = \beta$ the two cubes collapse into one, so $a(x - \alpha)^3 + b(x - \alpha)^3$ could only ever be $(x - \alpha)^3$, which has no x-term to match $-3px$. That is why we needed a different approach. Writing the cubic in powers of $(x - r)$ shows what happens instead: $x^3 - 3r^2x + 2r^3 = (x - r)^3 + 3r(x - r)^2$, so the double root sits at the repeated root $t = r$ of the quadratic.

Checks

- The real root is $2(2 - 1.259921)/(1 - 1.259921) \approx -5.694644$, and substituting gives $x^3 - 24x + 48 \approx 0$ to about 10^{-13}. The other two are $2.847322 \pm 0.567212\,\mathrm{i}$, also with zero residual.
- The three roots sum to 0 (no x^2 term) and multiply to -48: $-5.694644 \times (2.847322^2 + 0.567212^2) \approx -48.0$.
- One real root and two complex ones fits the theory: $x^3 - 3px + q$ has three real roots only when $q^2 < 4p^3$, and here $q^2 > 4p^3$.
- The identity can be tested numerically: we checked it at random complex x for 2000 random pairs (p, q), and the factorisation $(x - r)^2(x + 2r)$ for several values of r.

Common slips

- Taking only the real cube root of $\frac{1}{2}$ and so finding only one solution. A cubic has three roots, and they come from the three cube roots.

- Mixing up which of a, b goes with which root: a (the coefficient of $(x-\alpha)^3$) has β on top.
- Swapping ω and ω^2 when clearing fractions. Since the two answers form a conjugate pair, the final set of roots is the same either way.

Beyond STEP

This is a version of the 16th-century method for solving cubics, often credited to Cardano and Tartaglia: reduce to a "depressed" cubic with no x^2 term, then solve an auxiliary quadratic (the *resolvent*). Galois theory explains why cubics and quartics have such resolvents and why general quintics do not. Allenby, *Rings, Fields and Groups*, gives a readable route towards these ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.